# 🚦 Linux Pipe (`|`) Command Cheat Sheet

The **pipe (`|`)** in Linux is used to **connect the output of one command to the input of another command**. It allows combining multiple commands to create powerful command chains.

👉General syntax:

```
command1 | command2 | command3 ...
```

- `command1` → generates output (stdout). - `|` → sends that output as input (stdin) to the next command. - `command2` → processes the input.

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## 🗾Basic Examples

### 1. Combine `ls` with `more` / `less`

```
$ ls -l | more
```

Shows file list page by page.

```
$ ls -l | less
```

Better pager (can scroll forward & backward).

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### 2. Search Inside Output with `grep`

```
$ ps aux | grep firefox
```

Shows only the processes containing `firefox`.

```
$ dmesg | grep error
```

Filters kernel logs for `error` messages.

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### 3. Count Results with `wc`

```
$ ls | wc -l
```

Counts the number of files in the current directory.

```
$ cat file.txt | wc -w
```

Counts the number of words in a file.

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### 4. Sort and Remove Duplicates

```
$ cat sample.txt | sort | uniq
```

Sorts the lines and removes duplicates.

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### 5. Chain Multiple Commands

```
$ cat logfile.log | grep "ERROR" | sort | uniq -c | sort -nr
```

Step by step: 1. `cat logfile.log` → read log file. 2. `grep "ERROR"` → filter lines with `ERROR`. 3. `sort` → sort results. 4. `uniq -c` → count duplicate lines. 5. `sort -nr` → sort by number (descending).

Output → shows most frequent errors.

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### 6. Monitor Logs in Real-Time

```
$ tail -f /var/log/syslog | grep "network"
```

Continuously prints log lines that contain `network`.

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### 7. Disk Usage Example

```
$ du -sh * | sort -h
```

Shows size of each item in directory, sorted by size.

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## 🏠Summary

- `|` connects commands together.

- Each command works on the output of the previous one.
- Useful for **filtering, counting, sorting, searching, and chaining multiple commands**.

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👈 Pipes make Linux commands more powerful by allowing you to build **mini data-processing pipelines** on the fly.